Milestone M3 (Lane B): Code-to-Report Audit & Discrepancy Log

Project: IIT/SRIC/R/AEH/2026/104 — 2-DOF Electric Cable Shovel Dipper–Handle Digital Twin
Audit Version: v1.0.0 \rightarrow v2.0.0-PRELIMINARY
Validation Status: PRELIMINARY (Pending frozen MS02P05 / 05_configuration_sweep Simulink execution)

1. Executive Summary of Audit Findings

A line-by-line code-to-report audit was conducted across the initial Milestone M3 MATLAB script (m3_observer_suite.m v1.0), the numerical experiment runner (run_m3_experiments.py v1.0), the initial technical report (M3_LaneB_Momentum_Observer_Report.pdf v1.0), and the primary repository files in shovel-dynamics_m2 (shovel_params.m, recover_load.m, M2_null.mat, M2_100kN_0p1111Hz_sp1p50.mat, and stage1_results.mat).
All seven audit issues identified in the review have been resolved in **m3_observer_suite.m (v2.0.0-PRELIMINARY)** and **M3_LaneB_Momentum_Observer_Report.pdf (v2.0.0-PRELIMINARY)**:

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
AUD-01	**Data Ingestion & Validation**	build_reference_trajectory(mat_filepath, P) in m3_observer_suite.m v1.0 accepted mat_filepath (defaulting to 'stage1_results.mat'), never called load(mat_filepath), and reconstructed a 28 s piecewise step trajectory (d_4 \in [2.6, 4.1]\ \mathrm{m}). Meanwhile, stage1_results.mat (1{,}020\ \mathrm{bytes}) is an M1 summary struct (results.ident, results.params_used) containing zero time-series vectors, while run_m3_experiments.py v1.0 fell back to an	Replaced with load_and_validate_mat_archive(filepath, P) in m3_observer_suite.m and a native HDF5 v1 binary reader (MatV73Reader) in run_m3_experiments_v2.py that directly ingests M2_null.mat (SHA-256: 74a2ed3f...) and M2_100kN_0p1111Hz_sp1p50.mat (SHA-256: 6050507d...). Validates all 20 R.sig arrays, units, monotonicity, uniform 100\ \mathrm{Hz} sampling, and parameter lock against shovel_params.m. Passing stage1_results.mat or

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		analytical sinusoid when scipy.io.loadmat failed on MATLAB v7.3 HDF5 files (M2_null.mat, M2_100kN_0p1111Hz_sp1p50.mat).	any missing/incomplete file now fails immediately with m3_observer_suite:SummaryOnlyMatFile / FileNotFound. Synthetic trajectory generation is isolated to opt-in 'Mode', 'synthetic_unit_test'.
AUD-02	**Sign Convention & Load Channel Mapping**	Report v1.0 §1.2 stated \tau_L = -Q_{\mathrm{ext}} = e_\tau = \text{ref}, whereas comparing D\ddot{q} + C\dot{q} + G + F_v\dot{q} + F_c = \tau_a + \tau_L against recover_load.m line 34 (D\ddot{q} + C\dot{q} + G + f = \tau_{\mathrm{actuator}} + Q_{\mathrm{ext}}) requires \tau_L = +Q_{\mathrm{ext}} = +J_s(q)^T F = [\text{R.sig.Qq3_pred}; \ \text{R.sig.Qd4_pred}] = -e_\tau = -\text{ref}. Furthermore, v1.0 used synthetic uncoupled sinusoids (245\ \mathrm{kN\cdot m}, 49\ \mathrm{kN}) instead of the actual Jacobian-coupled loads in M2_100kN_0p1111Hz_sp1p50.mat (peak 734.24\ \mathrm{kN\cdot m} and 86.52\ \mathrm{kN}).	Corrected sign reconciliation in §1.2 and code so \tau_L \equiv +Q_{\mathrm{ext}} = [\text{R.sig.Qq3_pred}; \ \text{R.sig.Qd4_pred}] = -e_\tau = -[\text{R.sig.ref3};\ \text{R.sig.ref4}]. Also added **Check 0 (run_raw_slx_diagnostic)** documenting that P.model_has_coulomb = false in shovel_params.m line 50 during the 2026-08-25 Simulink runs.
AUD-03	**Report/Code**	Report v1.0 tables	Implemented both

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	Consistency (B1 vs. B2)	listed both B1: Constant (Reversing) and B2: Constant (Unidirectional), but m3_observer_suite.m v1.0 only implemented a single 'B_Constant' entry in cases.	B1_Constant_Reversing ($t \in [5.0, 30.0]\ \mathrm{s}$, $N_{\mathrm{eval}} = 751$, 3 velocity reversals at t = 5.10, 15.10, 25.10 s) and B2_Constant_Unidirectional ($t \in [15.5, 24.5]\ \mathrm{s}$, $N_{\mathrm{eval}} = 271$, strictly $\dot{q}_3 > 0$, $\dot{d}_4 > 0$, 0 reversals) explicitly in m3_observer_suite.m and run_m3_experiments_v2.py. All report tables and figures are generated directly from the saved output (m3_audit_results.json / M3_observer_audit_results.mat).
AUD-04	**Noise Provenance (Test Case G)**	Report v1.0 and code v1.0 referred to Test G as "G_Noisy (Sensor + Act. Noise)", implying sensor-calibrated noise levels when no sensor noise specifications exist in the repository.	Renamed and relabeled everywhere as **G_Synthetic_Noise_Stress (Uncalibrated Synthetic Gaussian Noise Stress Test)**, using a portable deterministic Park–Miller LCG + Box–Muller generator (seed = 202603) that produces identical realizations in MATLAB and Python.
AUD-05	**Branch B Uncertainty Bound Separation**	m3_observer_suite.m v1.0 printed only a single CovLag[%] column while Report	Explicitly decomposed Branch B total error $\tau_L - \hat{\tau}_{L,B}$ into (1) filtered friction

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		v1.0 mixed raw and lag-compensated coverage in prose/tables.	discrepancy \hat{\tau}_{L,A} - \hat{\tau}_{L,B} (\vert{}\hat{F}_{c,\mathrm{filt},i}\vert{} \le f_{ci}), (2) FOH inter-sample chord error \tau_{L,\mathrm{filt}} - \hat{\tau}_{L,A}, and (3) dynamic observer tracking lag \tau_L - \tau_{L,\mathrm{filt}}. Both code and report now output three separate coverage columns: CovFricPct, CovLagPct, and CovRawPct.
AUD-06	**Task-Space Jacobian Factorization & Station Offset**	Report v1.0 §6 wrote J_P(q) = R(q_3)\,\mathrm{diag}(r_P, 1) and J_P(q)^{-T} = R(q_3)\,\mathrm{diag}(1/r_P, 1), omitting the column-swap permutation matrix \Pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} required because q = [q_3, d_4]^T orders the transverse revolute coordinate first and radial prismatic coordinate second. Also, m3_observer_suite.m v1.0 used r_bail = d4 + 2.05 m over d_4 \in [2.5, 4.5]\ \mathrm{m} instead of s_bail = d4 + 1.50 m over d_4 \in [6.75, 10.75]\ \mathrm{m} from recover_load.m line 17.	Corrected factorization to J_s(q) = Q(q_3)\,\mathrm{diag}(s(d_4), 1) = R(q_3)\Pi\,\mathrm{diag}(s(d_4), 1) with \det(J_s) = -s(d_4), J_s(q)^{-T} = R(q_3)\Pi\,\mathrm{diag}(1/s(d_4), 1), and F_{\mathrm{equiv}} = \frac{\tau_3}{s}\hat{e}_\theta(q_3) + F_4\hat{e}_r(q_3) = R(q_3)[F_4;\ \tau_3/s]. Verified numerically against R.sig.Qq3_pred, R.sig.Qd4_pred, R.sig.Fx, and R.sig.Fy in M2_100kN_0p1111Hz_sp1p50.mat (s_{\mathrm{bail}} = d_4 + 1.50\ \mathrm{m} \in [8.2505, 12.2495]\ \mathrm{m}, inverse

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			recovery error \le 2.91\times 10^{-11}\ \mathrm{N}).
AUD-07	**Reproducibility & PRELIMINARY Lock**	v1.0 lacked archive SHA-256 hashes, .slx provenance metadata, and portable RNG parity between MATLAB and Python.	Recorded full archive provenance (filename, SHA-256, R.prov.timestamp, R.prov.gitCommit, R.prov.matlab, N_raw=3001, fs_raw=100 Hz, TrimStart=5.0 s, k_frozen=20 s^-1, M=67, seed=202603, code_version=2.0.0-PRELIMINARY) and maintained **PRELIMINARY** status across all outputs pending frozen MS02P05 signals.

2. Quantitative Shift Log (v1.0 Synthetic vs. v2.0.0-PRELIMINARY Actual Archive Ingestion)

Ingesting the actual M2_null.mat and M2_100kN_0p1111Hz_sp1p50.mat trajectories (d_4 \in [6.7505, 10.7495]\ \mathrm{m}, q_3 \in [30^\circ, 60^\circ], t \in [5.0, 30.0]\ \mathrm{s}, N_{\mathrm{eval}} = 751 at 30\ \mathrm{Hz}) produced the following verified shifts relative to v1.0:

- **Test Case A (A_Zero_M2_null, direct M2_null.mat)**:
  - Saddle Torque (\tau_{L,3}) RMSE shifted from 55.59\ \mathrm{N\cdot m} (v1.0) to 53.47\ \mathrm{N\cdot m} (Branch A) and from 297.93\ \mathrm{N\cdot m} to 300.43\ \mathrm{N\cdot m} (Branch B).
  - Crowd Force (F_{L,4}) RMSE shifted from 4.24\ \mathrm{N} (v1.0) to 4.36\ \mathrm{N} (Branch A) and from 473.52\ \mathrm{N} to 472.84\ \mathrm{N} (Branch B).
  - In M2_null.mat, the velocity zero-crossings occur at t = 5.10, 15.10, 25.10\ \mathrm{s} (which coincide with exact 30\ \mathrm{Hz} sample nodes k = 153, 453, 753 where \vert{}\dot{q}_3\vert{} = 2.12\times 10^{-7}\ \mathrm{rad/s} \ll \epsilon_3 and \vert{}\dot{d}_4\vert{} = 1.62\times 10^{-6}\ \mathrm{m/s} \ll \epsilon_4), reducing the zero-load during-reversal peak in Branch A from 38.85\ \mathrm{N\cdot m} (caused by the artificial phase shift in v1.0) to 7.10\ \mathrm{N\cdot m} (q_3) and 0.26\ \mathrm{N} (d_4).
- **Test Cases B1 & B2 (B1_Constant_Reversing vs. B2_Constant_Unidirectional on M2_null.mat)**:

    - Over the full reversing window $t \in [5.0, 30.0]\ \mathrm{s}$ (B1), sign reversals partially cancel Branch B's unmodeled friction, giving a mean bias of $+51.23\ \mathrm{N\cdot m}$ (q_3) and $+89.36\ \mathrm{N}$ (d_4) alongside an MAE of $297.62\ \mathrm{N\cdot m}$ and $471.62\ \mathrm{N}$.
    - Over the strictly unidirectional window $t \in [15.5, 24.5]\ \mathrm{s}$ (B2, $N_{\mathrm{eval}} = 271$, $\dot{q}_3 > 0$, $\dot{d}_4 > 0$), Branch B's mean bias is exposed as $-304.10\ \mathrm{N\cdot m}$ (q_3, comprising $-340.00\ \mathrm{N\cdot m}$ Coulomb friction $+35.90\ \mathrm{N\cdot m}$ dynamic FOH chord offset) and $-475.69\ \mathrm{N}$ (d_4, comprising $-480.00\ \mathrm{N}$ Coulomb friction $+4.31\ \mathrm{N}$ dynamic FOH chord offset), compared to $+35.90\ \mathrm{N\cdot m}$ and $+4.31\ \mathrm{N}$ in Branch A.
- **Test Case C (C_Sinusoidal_M2_100kN, direct M2_100kN_0p1111Hz_sp1p50.mat)**:
    - Because M2_100kN_0p1111Hz_sp1p50.mat applies $F_y(t) = 100\ \mathrm{kN}\sin(2\pi t/9)$ at the bail station $s(t) = d_4(t) + 1.50\ \mathrm{m} \in [8.2505, 12.2495]\ \mathrm{m}$, the true generalized loads $Q_{q_3,\mathrm{pred}}(t) = s(t)F_y(t)\cos q_3(t)$ and $Q_{d_4,\mathrm{pred}}(t) = F_y(t)\sin q_3(t)$ have peaks of $734.24\ \mathrm{kN\cdot m}$ and $86.52\ \mathrm{kN}$ (vs. the synthetic $245\ \mathrm{kN\cdot m}$ and $49\ \mathrm{kN}$ in v1.0).
    - Consequently, at $k_3 = k_4 = 20\ \mathrm{s}^{-1}$, Branch A's total tracking RMSE is $16{,}694.23\ \mathrm{N\cdot m}$ (2.27% of peak, phase lag $50.05\ \mathrm{ms} = 2.00^\circ$, amplitude error -0.065%) on q_3 and $1{,}766.46\ \mathrm{N}$ (2.04% of peak, phase lag $50.05\ \mathrm{ms}$, amplitude error -0.070%) on d_4.
- **Test Case G (G_Synthetic_Noise_Stress on M2_100kN_0p1111Hz_sp1p50.mat)**:
    - Because $d_4 \in [6.75, 10.75]\ \mathrm{m}$ in M2_100kN_0p1111Hz_sp1p50.mat yields $D_{11}(d_4) = I_{zz} + M_d(d_4 - 1.32)^2 \in [1.89\times 10^6,\ 5.12\times 10^6]\ \mathrm{kg\cdot m^2}$ (vs. $\sim 5.5\times 10^5\ \mathrm{kg\cdot m^2}$ when $d_4 \in [2.6, 4.1]\ \mathrm{m}$), the uncalibrated synthetic velocity perturbation $\sigma_{\dot{q}_3} = 3\ \mathrm{mrad/s}$ amplified by $k_3 D_{11}(d_4)$ produces a saddle RMSE of $180.05\ \mathrm{kN\cdot m}$ at $k = 20\ \mathrm{s}^{-1}$ (and $4.74\ \mathrm{kN}$ on d_4 where $D_{22} = M_d = 54{,}300\ \mathrm{kg}$ is constant).
- **Task-Space Jacobian (run_jacobian_analysis on M2_100kN_0p1111Hz_sp1p50.mat)**:
    - Across the actual archive stroke $d_4 \in [6.7505, 10.7495]\ \mathrm{m}$, the COG lever arm is $s_{\mathrm{COG}} = d_4 - 1.32 \in [5.4305, 9.4295]\ \mathrm{m}$ ($\kappa_2(J_{\mathrm{COG}}) \in [5.43, 9.43]$) and the bail lever arm is $s_{\mathrm{bail}} = d_4 + 1.50 \in [8.2505, 12.2495]\ \mathrm{m}$ ($\kappa_2(J_{\mathrm{bail}}) \in [8.25, 12.25]$), yielding an equivalent bail tip-force Coulomb friction floor $\Vert{}J_{\mathrm{bail}}^{-T} F_c\Vert{}_2 = \sqrt{(340/s_{\mathrm{bail}})^2 + 480^2} \in [480.80,\ 481.77]\ \mathrm{N}$.